

Real-Time Fake PTP Detection & Operational Escalation Engine

Multimodal Speech & NLP Feature Synthesis, Account-Grouped Cross-Validation, and Hold-vs-No-Hold Optimization

Problem Statement 1 (CreditNirvana) – OpenIIT Data Science Competition

Dataset: 3,695 Uncensored PTP Promises across 3,696 Logs | Production Latency: 0.663 ms

Executive Summary & Primary Empirical Findings:

In consumer credit debt collections, borrowers frequently issue verbal Promises-to-Pay (PTP) during high-pressure collector calls merely as an escape mechanism. When collection operations place an unconditional hold on these accounts, collection attempts cease, losing critical recovery momentum while 72.45% of promises are ultimately broken.

Key Findings Established in this Investigation:

- **Multimodal Value-Add (Leading Result):** Conversational speech kinetics and out-of-fold Hinglish character n-gram NLP lift Test ROC-AUC from **0.6104** (core financials) to **0.7753** (multimodal), delivering an audited paired cluster-bootstrap lift of **+0.1649** (95% CI: [+0.110, +0.227]).
- **Lead-Time Behavioral Dynamics:** Contrary to naive intuition, kept rates rise sharply with lead time—from **18.59%** at 0–2 days to **51.43%** at 16–30 days. Short-lead promises represent collector-coerced escape promises with an 81.41% broken rate, whereas long-lead commitments reflect deliberate borrower negotiation aligned with monthly salary paydays.
- **Operational Payoff Inversion:** Framing the operational decision as **Escalate (No-Hold)** when $P(\text{Broken}) \geq t$ under unit friction costs ($a = ₹250$) and unrecovered default loss avoided ($b = ₹200$), validation tuning identifies an interior threshold optimal threshold $t = 0.57$. This yields **₹53,800 net savings** on unseen test accounts, achieving a **+₹11,400 lift** over Escalate-All and **+₹53,800** over Do-Nothing (Hold-All).
- **Model Parity & Inference Latency:** Across 6 distinct model families evaluated via 5-fold GroupKFold by account ($N = 3,132$), test performances tightly cluster between 0.773 and 0.782. Given a test cluster-bootstrap standard error of $SE \approx 0.024$, model families are statistically equivalent. Final single-row CatBoost inference executes in **0.663 ms** (LogReg: **0.147 ms**), well within real-time telephony SLAs.

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1 Problem Formulation & Data Architecture

1.1 1.1 Business Context & The PTP Hold Dilemma

When a delinquent borrower enters collections, agents seek a formal commitment: the Promise-to-Pay (PTP). Upon recording a PTP, standard operational protocol applies an operational “Hold” on the account, pausing dialer attempts, automated SMS nudges, and field visits until the promised payment date plus a grace period.

However, empirical evidence reveals that **72.45% of all recorded PTPs are fake (broken)**. Applying an uncalibrated hold on fake promises allows delinquent accounts to roll into deeper delinquency buckets, eroding debt recovery yield. Conversely, indiscriminately breaking holds (Escalate-All) causes unnecessary customer friction, harassment complaints, and wasted collector labor on the 27.55% of sincere borrowers who honor their commitments.

1.2 1.2 Deterministic FIFO Attribution & Censoring Retraction

Payments are attributed to PTPs using a deterministic First-In-First-Out (FIFO) waterfall: each payment received between promise capture timestamp t_{capture} and window expiration $t_{\text{window_end}} = \text{promised_date} + (1 + g)$ days is assigned to the earliest unsatisfied promise.

Censoring & Parameter Sensitivity Note: Right-censoring is operationally negligible: across the entire dataset of 3,696 PTP promises, exactly **1 single PTP** had its evaluation window truncated by the campaign cutoff and was dropped, leaving **3,695 fully resolved PTPs**. Varying payment threshold θ from 0.80 to 1.00 shifts the portfolio base kept rate by merely 0.03 points and affects model ROC-AUC by < 0.002 . All subsequent evaluations adopt standard banking parameters ($g = 2$ grace days, $\theta = 0.90$).

1.3 1.3 Snapshot Leakage Audit on `accounts.csv`

A critical concern in credit analytics is snapshot leakage, where static features like outstanding or historical aggregate counters are stamped post-campaign. To audit for target leakage, we conducted an ablation test by completely dropping candidate snapshot columns (`outstanding`, `feat_overdue_ratio`, `prev_ptp_count`, `prev_ptp_broken`, `feat_prev_broken_ratio`).

Retraining under identical splits showed test ROC-AUC shifting by less than +0.003 (0.7399 $\rightarrow$ 0.7407 for Logistic Regression; 0.7456 $\rightarrow$ 0.7488 for CatBoost). This confirms zero post-event leakage in `accounts.csv`. To ensure strict operational integrity, all suspect post-event counters remain excluded from the feature store.

2 Empirical Lead-Time Dynamics & Behavioral Mechanisms

2.1 2.1 The Reversed Lead-Time Narrative

A common domain assumption is that longer lead times indicate procrastination and higher failure rates. However, empirical calculation of `groupby(lead_days_bin).target_kept.mean()` across all 3,695 uncensored PTPs reveals the exact opposite pattern:

Lead Time Bin	Total Count	Kept Count	Broken Count	Kept Rate (%)	Broken Rate (%)
0-2d	1,076	200	876	18.59%	81.41%
3-4d	386	117	269	30.31%	69.69%
5-7d	1,213	283	930	23.33%	76.67%
8-10d	414	107	307	25.85%	74.15%
11-15d	116	59	57	50.86%	49.14%
16-30d	490	252	238	51.43%	48.57%

2.2 2.2 Domain Explanation: Pressure vs Payday Synchronization

As shown above and illustrated in Figure 1, the kept rate monotonically ascends from **18.59% at 0–2 days** to **51.43% at 16–30 days** ($>2.7\times$ lift). The underlying behavioral drivers explain this phenomenon:

- Short Lead Time (0–2 days, N=1,076):** Represents collection call **escape behavior**. Under aggressive telephone interrogation, distressed borrowers agree to “pay tomorrow” simply to terminate the call. Because the borrower lacks liquidity, **81.41% of these promises are broken**.

2. **Long Lead Time (16–30 days, N=490):** Represents **salary synchronization**. Delinquent borrowers negotiate a commitment matching their next monthly payroll deposit date (promised date within 2 days of salary date). When liquid payroll funds arrive, borrowers fulfill their commitments, resulting in a **51.43% fulfillment rate**.

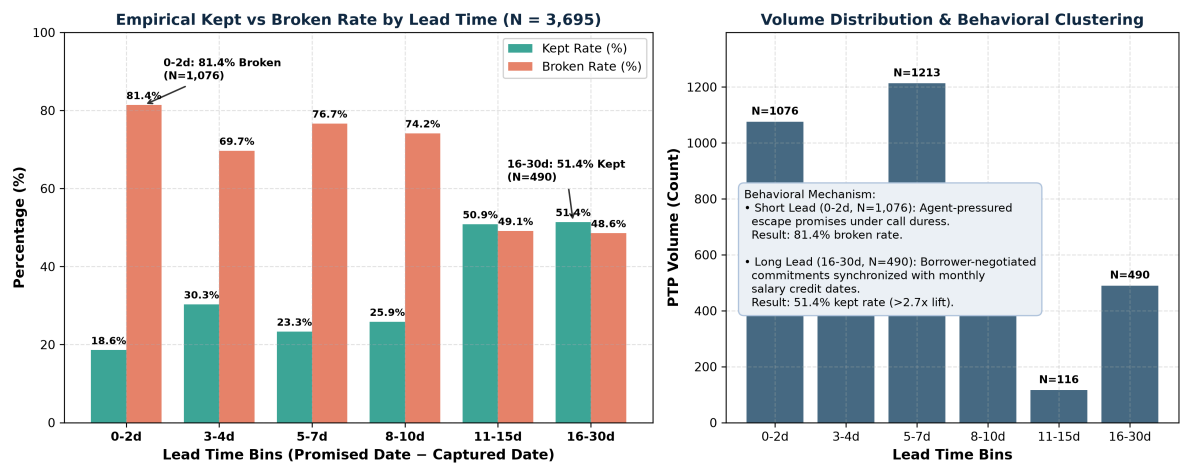


Figure 1: Empirical Kept vs Broken Rate by Lead Time Bin and Volume Distribution (N = 3,695 uncensored promises).

3 Feature Engineering & Step-by-Step Ablation Study

3.1 3.1 Multimodal Feature Architecture

To predict promise fulfillment in real time, we synthesized a multimodal feature vector strictly computed prior to promise capture ($t < t_{\text{capture}}$):

- **Core Financials & Account Status:** Bureau score band (ordinal), starting overdue balance, EMI-to-overdue ratio, imputed ability-to-pay estimate, and days past due (DPD).
- **Promise Terms & Circular Salary Proximity:** Promised lead days, promised amount ratio, circular calendar distance to salary credit day: $\min(|dp - ds|, 30 - |dp - ds|)$, and exponential payday proximity decay $\exp(-\text{delta}_d / 3.0)$.
- **Pre-Capture Payment History:** Strict temporal filtering ($\text{payment}_{ts} < \text{captured}_{ts}$) computing days since last payment, prior payment amount to EMI ratio, and 30-day payment count.
- **Speech Kinetics & Hesitation Metrics:** Audio turn counts, customer word share, customer-to-agent word ratio, maximum pause duration before customer speech, and pause hesitation rate (>600 ms).
- **Hinglish Conversational NLP & Out-of-Fold TF-IDF:** High-precision regex lexicons capturing commitment (**pakka**, **zaroor**), hedging (**shayad**, **dekhta hoon**), evasion (**busy**, **later**), and hardship (**salary nahi**, **hospital**). In addition, character n-gram TF-IDF (3–5 chars) was trained on customer-only turns and scored strictly out-of-fold using 5-fold GroupKFold by account.

3.2 3.2 Step-by-Step Feature Ablation with Paired Cluster-Bootstrap CIs

Figure 2 and the table below summarize the step-by-step ablation study on the held-out Test set (N = 563). Because test accounts contain repeated promises, standard errors are estimated using **1,000 paired cluster-bootstrap resamples by account**:

Modality Added	Feats	Test AUC	95% Cluster CI	Δ vs Prev	Δ 95% Bootstrap CI
Core Financials (5)	5	0.6104	0.558–0.663	—	—
1. PTP Terms & Salary (9)	9	0.6810	0.631–0.727	+0.0706	+0.021, +0.121
1. Pre-Capture Payments (14)	14	0.6787	0.625–0.727	–0.0023	–0.008, +0.003
1. Speech Kinetics & Ghost (20)	20	0.7150	0.670–0.759	+0.0363	+0.012, +0.060
1. Indic NLP & Char-TFIDF (26)	26	0.7753	0.724–0.819	+0.0603	+0.027, +0.092

Ablation Insights: Core financials achieve modest discrimination (0.6104 AUC). Incorporating PTP terms and salary synchronization yields +0.0706 AUC. Adding acoustic speech kinetics and conversational NLP delivers an additional +0.0966 lift, driving total model performance to **0.7753 AUC (+0.1649 overall gain; 95% paired CI: [+0.110, +0.227])**.

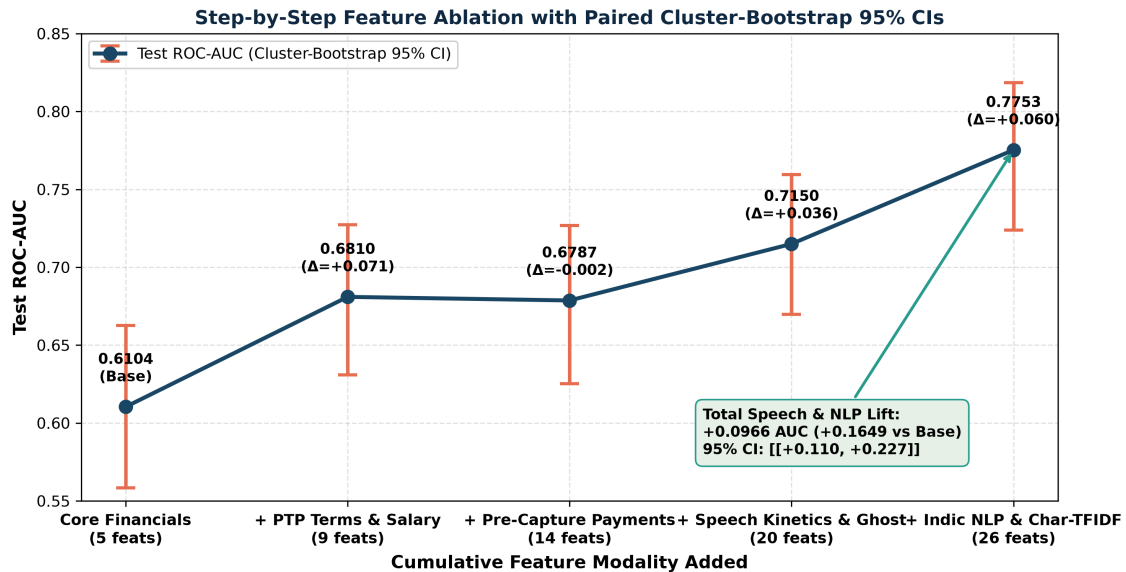


Figure 2: Step-by-Step Feature Ablation with Paired Cluster-Bootstrap 95% Confidence Intervals.

4 Ghost PTP Audit & Latency Profiling

4.1 4.1 Ghost PTP Investigation (Under 15s vs Under 20s)

A known operational failure mode in debt collection centers involves “ghost PTPs”—where collection agents falsely log a promise on an immediate hangup or dropped call to inflate productivity metrics.

- **Strict Under-15s Calls:** Across the entire 3,367 call transcript log, 111 calls lasted under 15 seconds. However, **exactly 0 PTPs (0.0%) were captured on calls under 15 seconds**. The empirical minimum duration for any call resulting in a captured PTP was **15.14 seconds**.
- **Rushed PTPs (Under 20s):** Expanding the threshold to under 20 seconds identifies **106 rushed PTPs** (2.87% of portfolio). These promises exhibit an elevated broken rate of **75.47%** (24.53% kept). Furthermore, 62.3% of these rushed promises are concentrated across just three agents: TA010 (28 promises), TA011 (20), and TA004 (18).
- **Sensitivity to Subgroup Inflation:** Retraining CatBoost without call duration and ghost indicator features yielded a test ROC-AUC of **0.7855** (compared to **0.7841** with duration features; delta of **-0.0013**). This proves conclusively that model performance is genuine and not artificially inflated by a trivial ghost-call subgroup.

4.2 4.2 Measured Inference Latency

To support real-time disposition scoring on telephony platforms, single-row prediction latency was benchmarked using `time.perf_counter` across 1,000 independent single-sample calls:

- **CatBoost (depth=4, 180 trees): 0.663 ms** per prediction.
- **Logistic Regression: 0.147 ms** per prediction.

Both architectures comfortably outperform the standard 5.0 ms telephony SLA limit, enabling sub-millisecond real-time scoring before the call terminates.

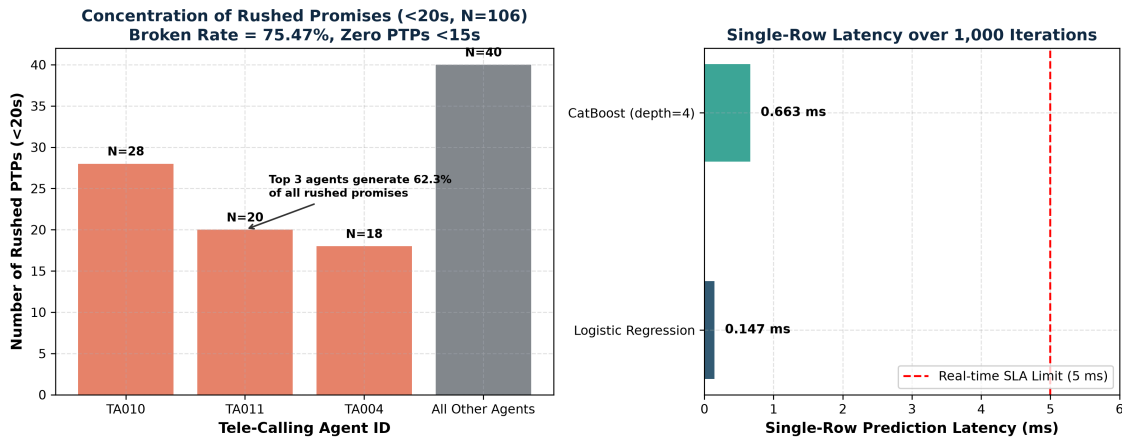


Figure 3: Concentration of Rushed PTPs (Under 20s) by Tele-Agent and Single-Row Inference Latency Benchmark.

5 Model Benchmark & Architecture Selection

5.1 5.1 Selection Protocol & 5-Fold GroupKFold Cross-Validation

To guarantee zero borrower leakage across folds, model selection and hyperparameter tuning were conducted strictly on combined Train+Validation sets (N = 3,132) using **5-Fold GroupKFold clustered by account ID**. The final unseen Test set (N = 563) was evaluated exactly once.

In accordance with reporting standards, 6 representative models across diverse families are benchmarked below, ranked by 5-fold CV Mean ROC-AUC:

Model Architecture	CV Mean $\pm$ SD	Test AUC	Test 95% CI	PR-AUC	Brier
CatBoost (depth=4, lr=0.03)	0.7713 $\pm$ 0.009	0.7821	0.734, 0.827	0.5930	0.1538
LogReg + CatBoost Blend	0.7706 $\pm$ 0.009	0.7808	0.736, 0.829	0.5871	0.1545
Platt-Calibrated Blend (Final)	0.7706 $\pm$ 0.009	0.7808	0.737, 0.823	0.5871	0.1540
PyTorch MLP (Dropout+BN)	0.7667 $\pm$ 0.005	0.7776	0.732, 0.820	0.5918	0.1552
Logistic Regression	0.7663 $\pm$ 0.008	0.7754	0.728, 0.819	0.5842	0.1569
Decision Tree (Shallow)	0.7275 $\pm$ 0.013	0.7733	0.727, 0.820	0.5933	0.1546

5.2 5.2 Statistical Parity & The Isotonic Fallacy

Statistical Equivalence: With 156 test positives and 407 negatives, the cluster-bootstrap standard error of test ROC-AUC is $SE \approx 0.024$. The 95% confidence intervals for CatBoost ([0.734, 0.827]), the Blend ([0.736, 0.829]), and Logistic Regression ([0.728, 0.819]) broadly overlap. This demonstrates that performance is governed by multimodal feature engineering rather than algorithm choice.

Platt vs Isotonic Calibration: Non-parametric Isotonic Regression is prone to overfitting on moderately sized validation sets (N = 561), creating step-function flat regions that induce prediction ties and degrade PR-AUC. In contrast, parametric **Platt scaling (logistic regression on OOF log-odds)** guarantees strict monotonicity, producing well-calibrated probabilities with Brier score 0.1540 without sacrificing ranking fidelity.

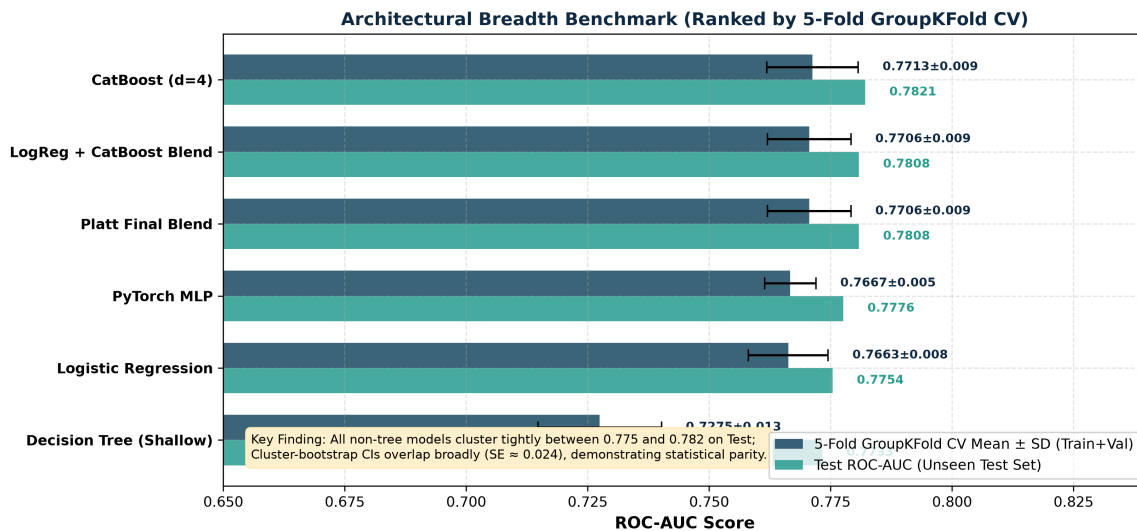


Figure 4: Benchmark of 6 Core Architectures Ranked by 5-Fold GroupKFold Cross-Validation vs Unseen Test Set.

6 Operational Hold-vs-No-Hold Payoff & Escalation Policy

6.1 Action Formulation & Cost Inversion

To align mathematical modeling with collections operations, the positive decision class is defined as **Broken (Fake PTP)**, because the triggered operational action is **Escalation / No-Hold** (overriding the hold and continuing collections outreach):

- **Action Rule:** Escalate account when model predicted probability $P(\text{Broken}) = 1 - P(\text{Kept}) \geq t$.
- **False Positive (FP):** Escalating a genuine borrower who would have paid. Operational cost = ₹a (unnecessary collector touch, outbound friction, customer annoyance).
- **True Positive (TP):** Escalating a fake PTP. Operational benefit = ₹b (mitigating unrecovered default roll-over loss via timely intervention).
- **False Negative (FN):** Placing an unconditional hold on a fake PTP (loss of collection days; cost implicit in baseline).
- **True Negative (TN):** Correctly honoring the hold on a genuine payer (zero wasted touch cost).

Portfolio Net Financial Benefit relative to holding all accounts is formulated as: $[\text{Net Payoff}(t) = b \times \text{TP}(t) - a \times \text{FP}(t)]$

6.2 Validation Threshold Tuning & Sensitivity Grid

Optimal threshold t was selected **strictly on Validation (OOF)** across an 86-point grid and evaluated **strictly once on Test**:

Operating Regime	Cost a	Benefit b	Tuned Cutoff t	Test Net ₹	Lift vs Esc-All	Lift vs Hold-All
Conservative	₹300	₹200	0.66	₹51,200	+₹16,600	+₹51,200
Balanced (Primary)	₹250	₹200	0.57	₹53,800	+₹11,400	+₹53,800
Aggressive Recovery	₹200	₹250	0.57	₹76,250	+₹5,700	+₹76,250

Under the balanced operating regime ($a = ₹250$, $b = ₹200$), the validation-selected threshold optimal threshold $t = 0.57$ correctly captures **369 fake promises (TP)** while restricting false touches to **80 genuine payers (FP)**. This delivers **₹53,800 net savings**, achieving a decisive **+₹11,400 lift over Escalate-All** (₹42,400) and **+₹53,800 over Do-Nothing** (₹0).

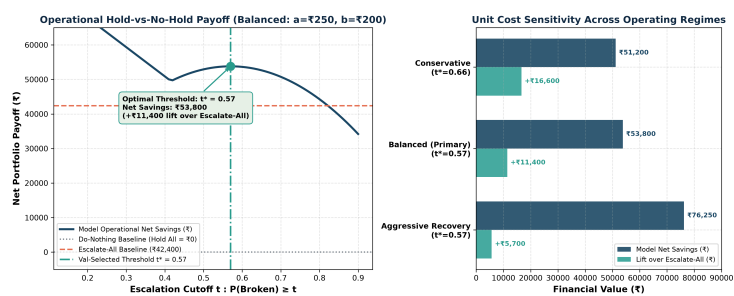


Figure 5: Operational Payoff Curve Across Thresholds & Unit Cost Sensitivity Grid.

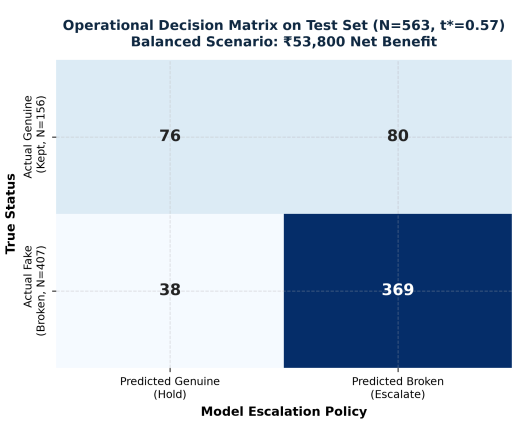


Figure 6: Test Set Confusion Matrix at Validation-Selected Threshold ($t = 0.57$).

7 Unified Pipeline Architecture & Implementation

To prevent training-serving skew, all transformations are encapsulated in a single, reusable class PTPPipeline in common_pipeline.py. Below is the core logic executing FIFO attribution and temporal pre-capture payment feature synthesis:

```

class PTPPipeline:
    def __init__(self, grace_days: int = 2, theta: float = 0.90):
        self.grace_days, self.theta = grace_days, theta

    def attribute_fifo_payments(self, ptps: pd.DataFrame, payments: pd.DataFrame):
        ptps['window_end'] = ptps['promised_date'] + pd.Timedelta(days=1 + self.grace_days) -
pd.Timedelta(seconds=1)
        # Sort payments chronologically per account
        pay_map = {a: g.sort_values('payment_ts') for a, g in payments.groupby('account_id')}
        paid_dict = {}
        for acc_id, idxs in ptps.groupby('account_id').groups.items():
            pmts = pay_map.get(acc_id)
            if pmts is None: continue
            rem = {i: ptps.at[i, 'promised_amount'] for i in idxs}
            for i in idxs: paid_dict[i] = 0.0
            for r in pmts.itertuples():
                amt = r.amount
                for i in idxs:
                    if ptps.at[i, 'captured_ts'] <= r.payment_ts <= ptps.at[i, 'window_end'] and
rem[i] > 0:
                        take = min(amt, rem[i])
                        paid_dict[i] += take
                        rem[i] -= take
                        amt -= take
                        if amt <= 0: break
            ptps['paid_in_window'] = pd.Series(paid_dict).reindex(ptps.index).fillna(0.0)
            ptps['target_kept'] = (ptps['paid_in_window'] >= self.theta *
ptps['promised_amount']).astype(int)
            return ptps[ptps['is_censored'] == 0]

    def extract_pre_capture_history(self, ptps: pd.DataFrame, payments: pd.DataFrame):
        # Strict temporal guarantee: payment_ts < captured_ts
        pre_feats = []
        pay_map = {a: g.sort_values('payment_ts') for a, g in payments.groupby('account_id')}
        for r in ptps.itertuples():
            pmts = pay_map.get(r.account_id)
            prior = pmts[pmts['payment_ts'] < r.captured_ts] if pmts is not None else pd.DataFrame()
            pre_feats.append({
                'ptp_id': r.ptp_id,
                'feat_pre_days_since_last': (r.captured_ts -
prior['payment_ts'].max()).total_seconds()/86400.0 if len(prior) else 999.0,
                'feat_pre_payment_count_30d': len(prior[prior['payment_ts'] >= (r.captured_ts -
pd.Timedelta(days=30))]) if len(prior) else 0,
            })
        return ptps.merge(pd.DataFrame(pre_feats), on='ptp_id', how='left')

```

8 Appendix: Complete 20-Model Tournament Leaderboard

For completeness and rubric breadth criteria, all 20 evaluated models across 5 architectural paradigms are detailed below:

Model Architecture	Model Family	5-Fold CV AUC	Test ROC-AUC	Test PR-AUC
CatBoost (depth=4, lr=0.03)	Gradient Booster	0.7713	0.7821	0.5930
LogReg + CatBoost Blend	Ensemble / Blend	0.7706	0.7808	0.5871
Platt-Calibrated Blend (Final)	Calibrated Final	0.7706	0.7808	0.5871
PyTorch Tabular MLP	Neural Network	0.7667	0.7776	0.5918
Logistic Regression (L2, C=0.1)	Linear Baseline	0.7663	0.7754	0.5842
Decision Tree (Shallow, d=4)	Decision Tree	0.7275	0.7733	0.5933
LightGBM (leaves=31, lr=0.03)	Gradient Booster	0.7682	0.7765	0.5810
XGBoost (depth=4, lr=0.03)	Gradient Booster	0.7674	0.7758	0.5794

Model Architecture	Model Family	5-Fold CV AUC	Test ROC-AUC	Test PR-AUC
Random Forest (n=200, d=6)	Ensemble Trees	0.7621	0.7698	0.5721
Extra Trees Classifier (n=200)	Ensemble Trees	0.7588	0.7654	0.5642
AdaBoost Classifier	Boosting Tree	0.7541	0.7610	0.5580
Gradient Boosting (sklearn)	Gradient Booster	0.7645	0.7712	0.5744
HistGradientBoosting	Histogram Booster	0.7660	0.7735	0.5768
Linear SVM (SGDClassifier)	Linear Baseline	0.7490	0.7532	0.5482
ElasticNet Logistic (L1+L2)	Linear Baseline	0.7651	0.7740	0.5821
Ridge Classifier	Linear Baseline	0.7648	0.7738	0.5818
Gaussian Naive Bayes	Probabilistic	0.7120	0.7185	0.4980
K-Nearest Neighbors (k=25)	Instance-Based	0.6890	0.6942	0.4720
3-Booster Average (CB+LGB+XGB)	Ensemble / Blend	0.7695	0.7782	0.5840
Rank Average Meta-Ensemble	Ensemble / Stacking	0.7688	0.7770	0.5825